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Dear Editors,

I am submitting “Likelihood approximations distort the ion channel kinetic information in macroscopic currents” for consideration as a Research Article.

We have reached the limit of what a macroscopic current shows by eye. The flip state we proposed for $P2X_2$ in 2007 was still on the visible side, a delay before the current rises, later confirmed. The questions now asked of these receptors, how subunits couple and whether they do so symmetrically, leave no such mark. They are settled by ratios of evidence between kinetic schemes, and whether the likelihoods behind those ratios report the information a recording actually holds had never been measured. This manuscript measures it, and finds that most of them do not.

We know the stake from the inside. Our 2025 report that $P2X_2$ activates asymmetrically rested on the evidence over nine schemes, and under a control likelihood differing only in its recursion a different mechanism came first and a factor of six in evidence disappeared. We published the sensitivity and argued for one arm. On a recording there is no truth to measure a likelihood against, and fitting well does not supply one: a recursive likelihood follows the record whether its assumptions hold or not.

So we went outside the fit. Channel gating simulates exactly where its likelihood cannot be evaluated exactly, and at known parameters a correct likelihood must satisfy two identities: its score averages to zero, and the covariance of that score equals the Fisher information it reports. We measured both for eight likelihoods, from least squares to a filter conditioning each interval average on both endpoints, over 210 combinations of channel number, noise and sampling interval, ten thousand exact recordings at each, in a two-state scheme so that the error measured is the approximation’s own.

Every likelihood that leaves the gating correlation unmodelled misreports its information, by up to an order of magnitude, and no rescaling repairs it because the distortion has directions. Our published control is one of them: across a record it accumulates about twenty times the information it holds, and the recursion is what brings that factor back to one. We do not recompute the $P2X_2$ evidences. We show that the arm favouring the alternative mechanism is distorted in the simplest model there is and the calibrated arm is not, and we audit that arm on the same grid, where it departs in the few-channel corner and we say by how much.

Münch and colleagues asked this of their own filter in eLife in 2022 by counting coverage, which tests a posterior and costs a fit per replicate. The identities cost one pass and return a matrix: which parameter is misreported, and whether the error is committed at each sample or accumulated across the record. We are not offering another filter; ours agrees to one part in 10^8 with an integrated-measurement Kalman filter known since 1988. The measurement is what is new.

For an experimentalist the map has one line that holds however the thresholds are drawn: no region gives both a trustworthy least-squares error bar and information about the unitary current, the two boundaries one to three decades of noise apart. The recordings rich enough to determine the amplitudes are the ones whose classical error bars are the wrong width. The calibrated likelihood, its score and Fisher information, the exact simulator and the diagnostic are a small library with R and Python bindings, so a reader can put the same two identities on their own scheme.

Yours sincerely,

Luciano Moffatt